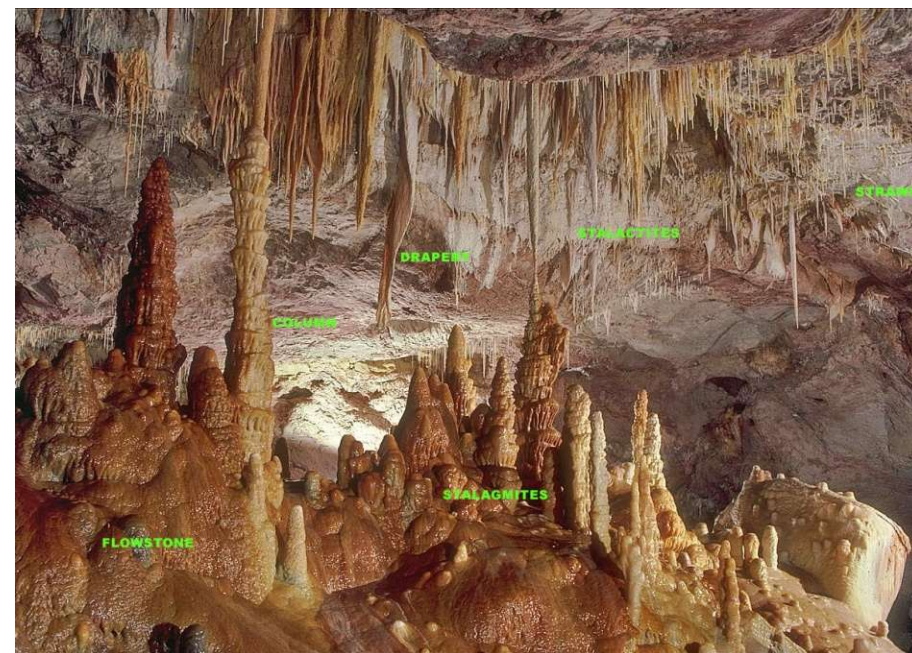




IDC 203: INTRODUCTION TO EARTH SCIENCES

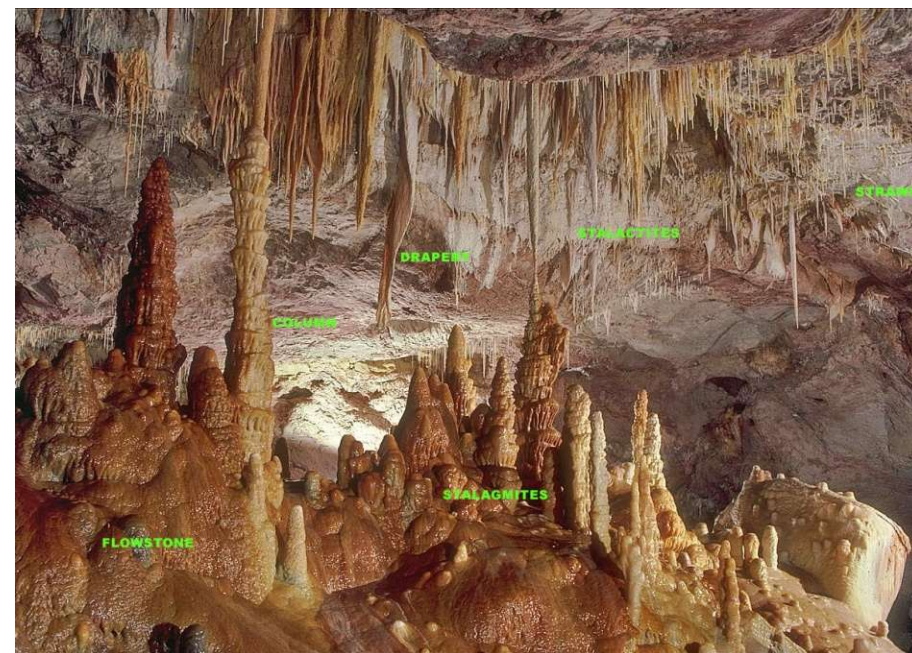


Introduction

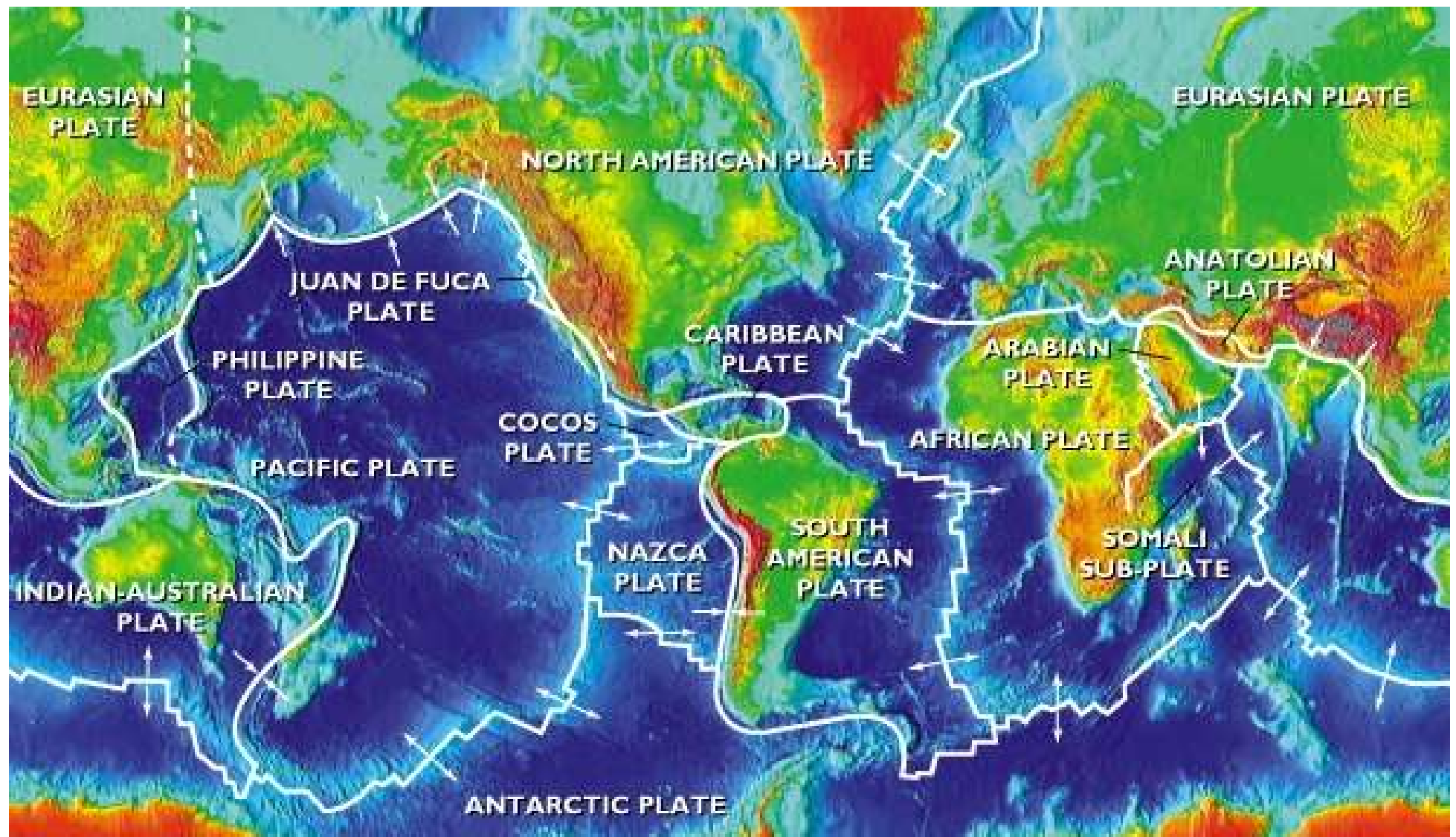
- **Plate tectonics**
- **Volcanoes**



IDC 203: INTRODUCTION TO EARTH SCIENCES



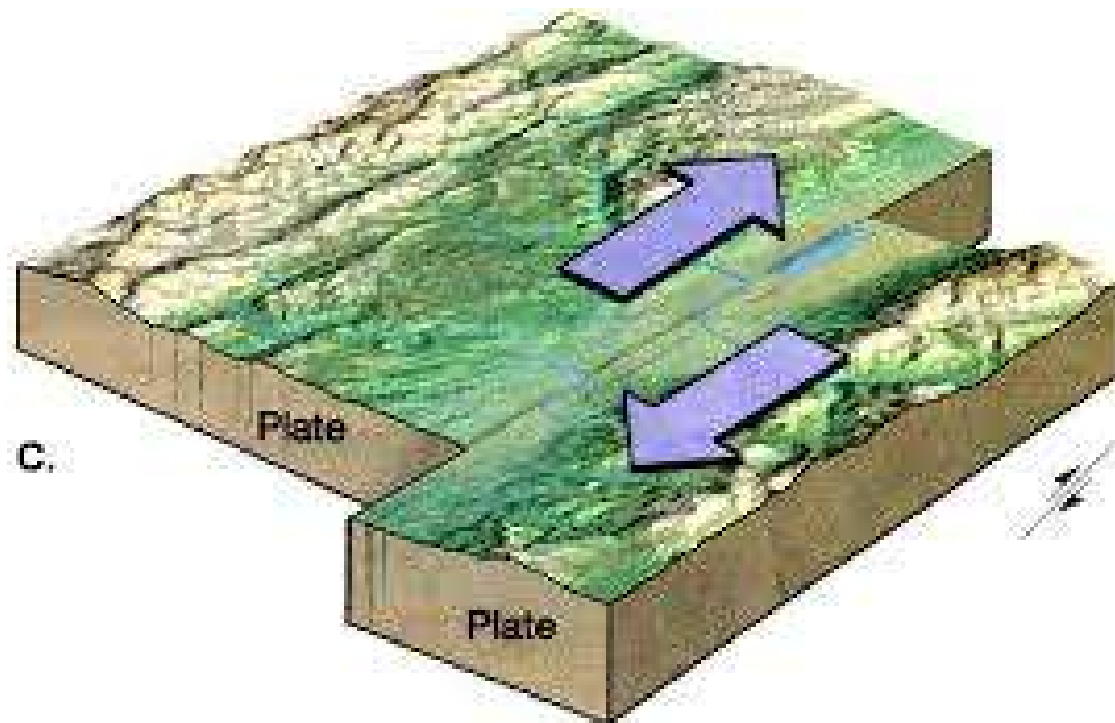
Present day plates



Transform plate boundary

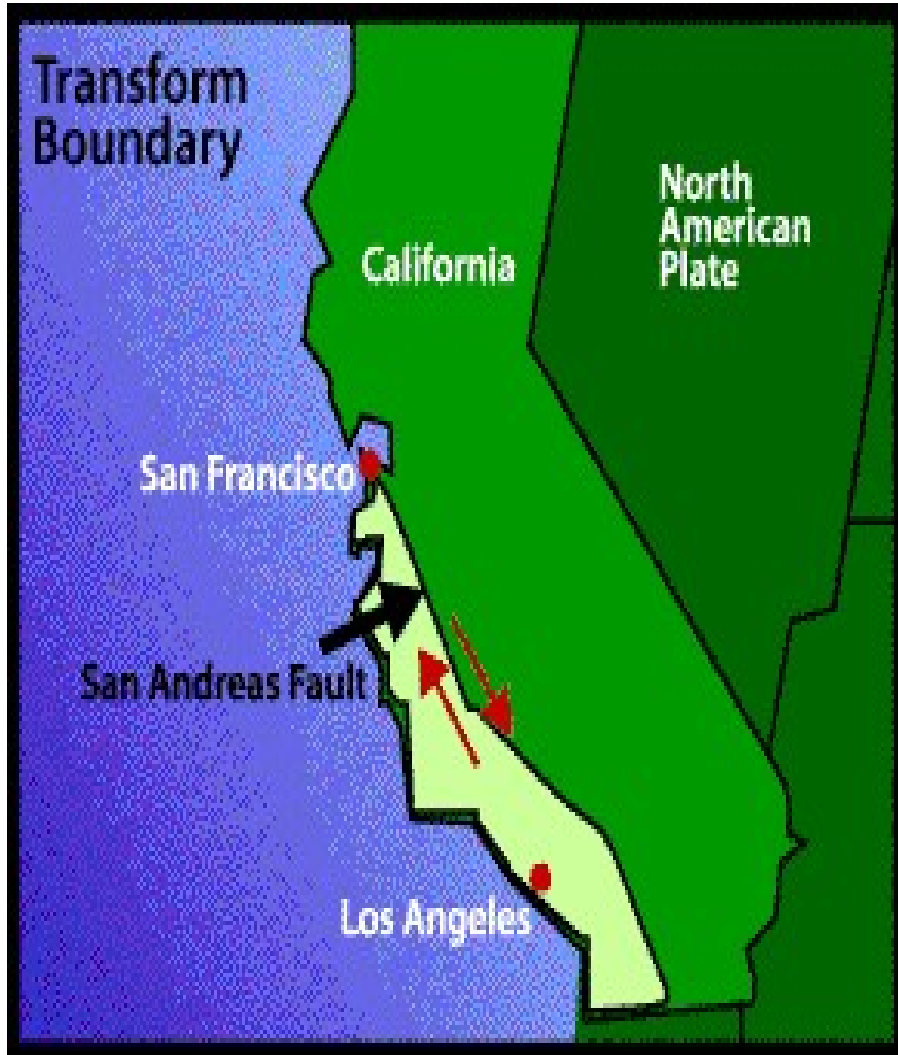
- Crust is neither created nor destroyed

Plates slide past one another



Transform plate boundary

San Andreas Fault



Carrizo Plains, Central California

Rates and History of plate movements

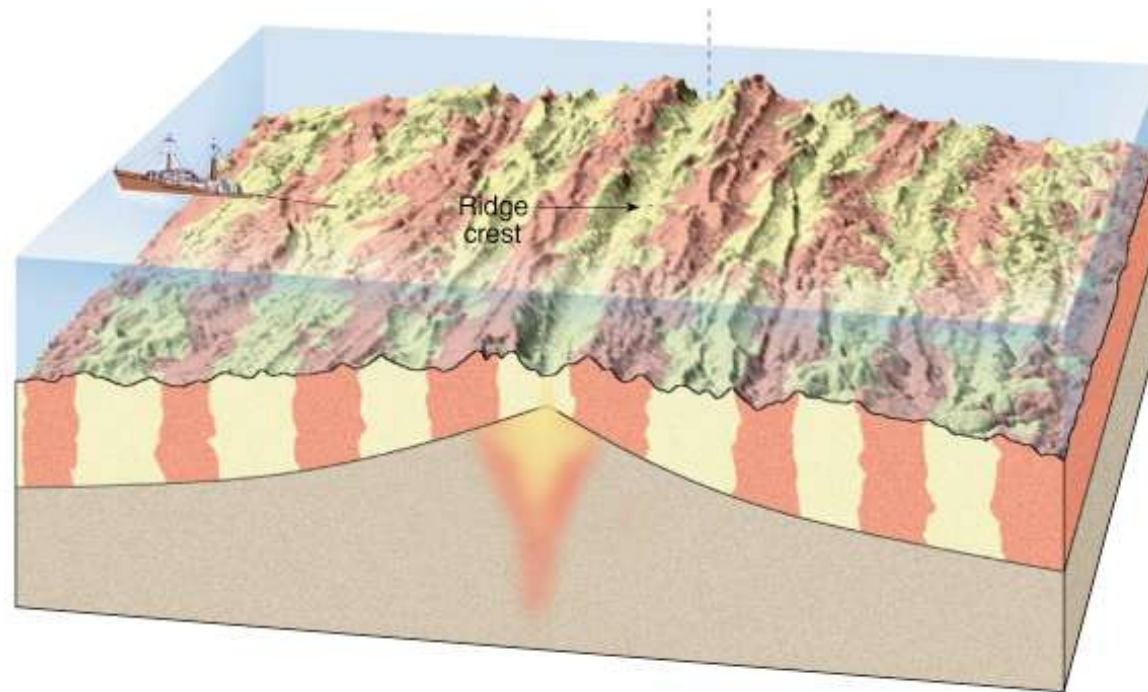
How fast do plates move?

Do some plates move faster than others, and if so, why?

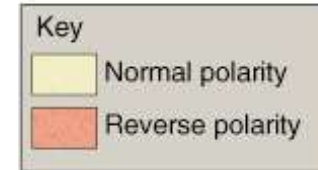
Is the velocity of plate movements today the same as it was in the Geologic past?

Rates and History of plate movements

Paleomagnetism

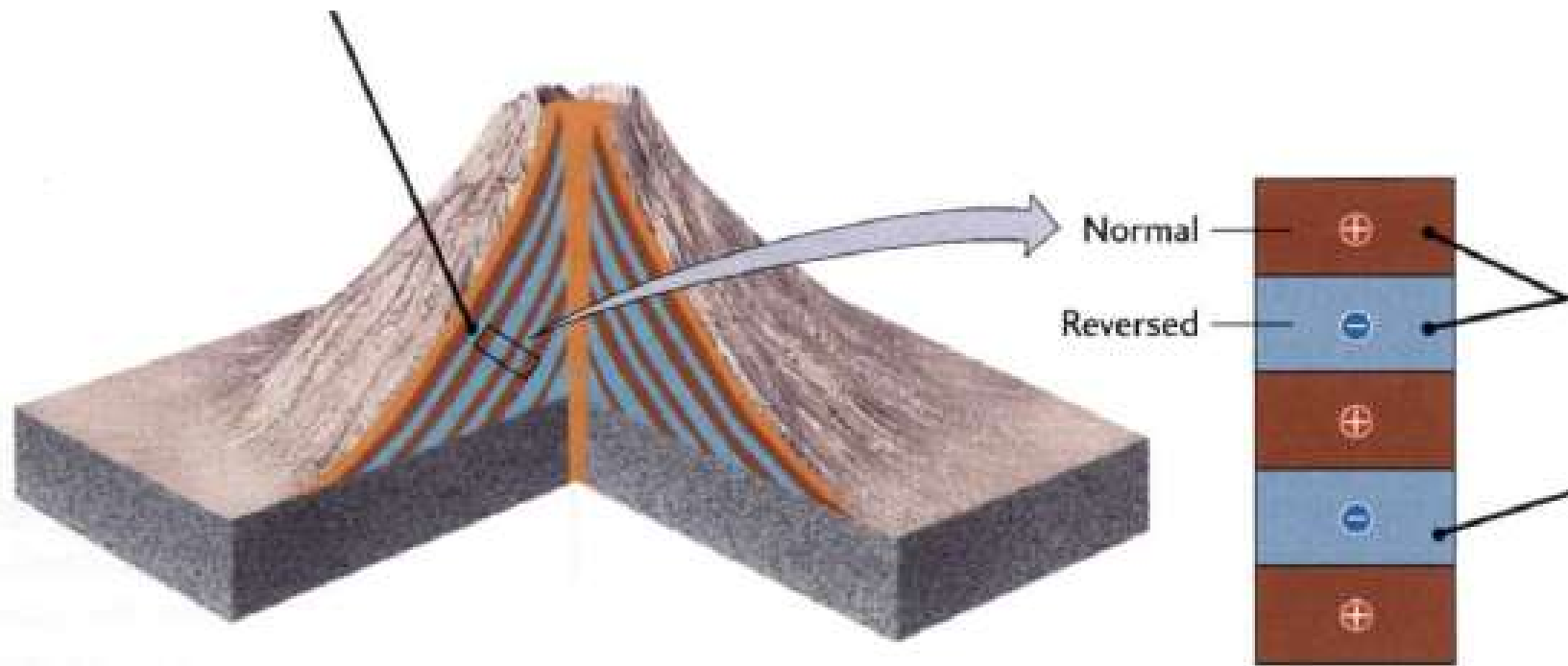


B. Research vessel towing magnetometer across ridge crest

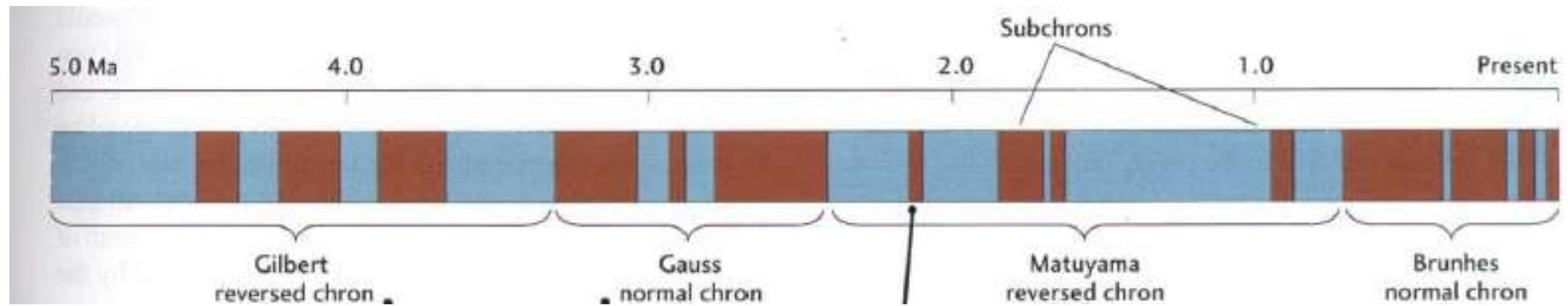


C. Location map

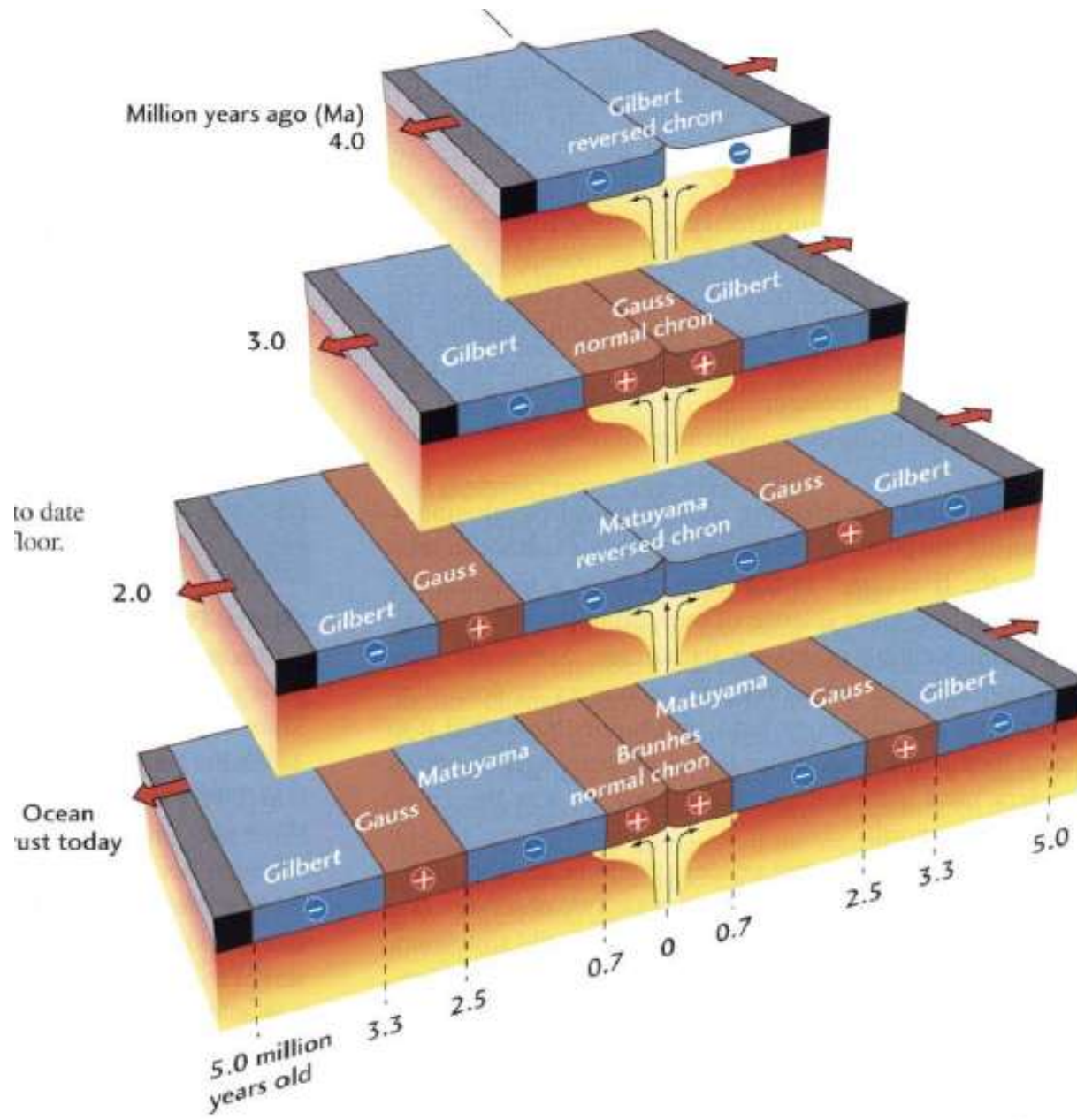
Paleomagnetism



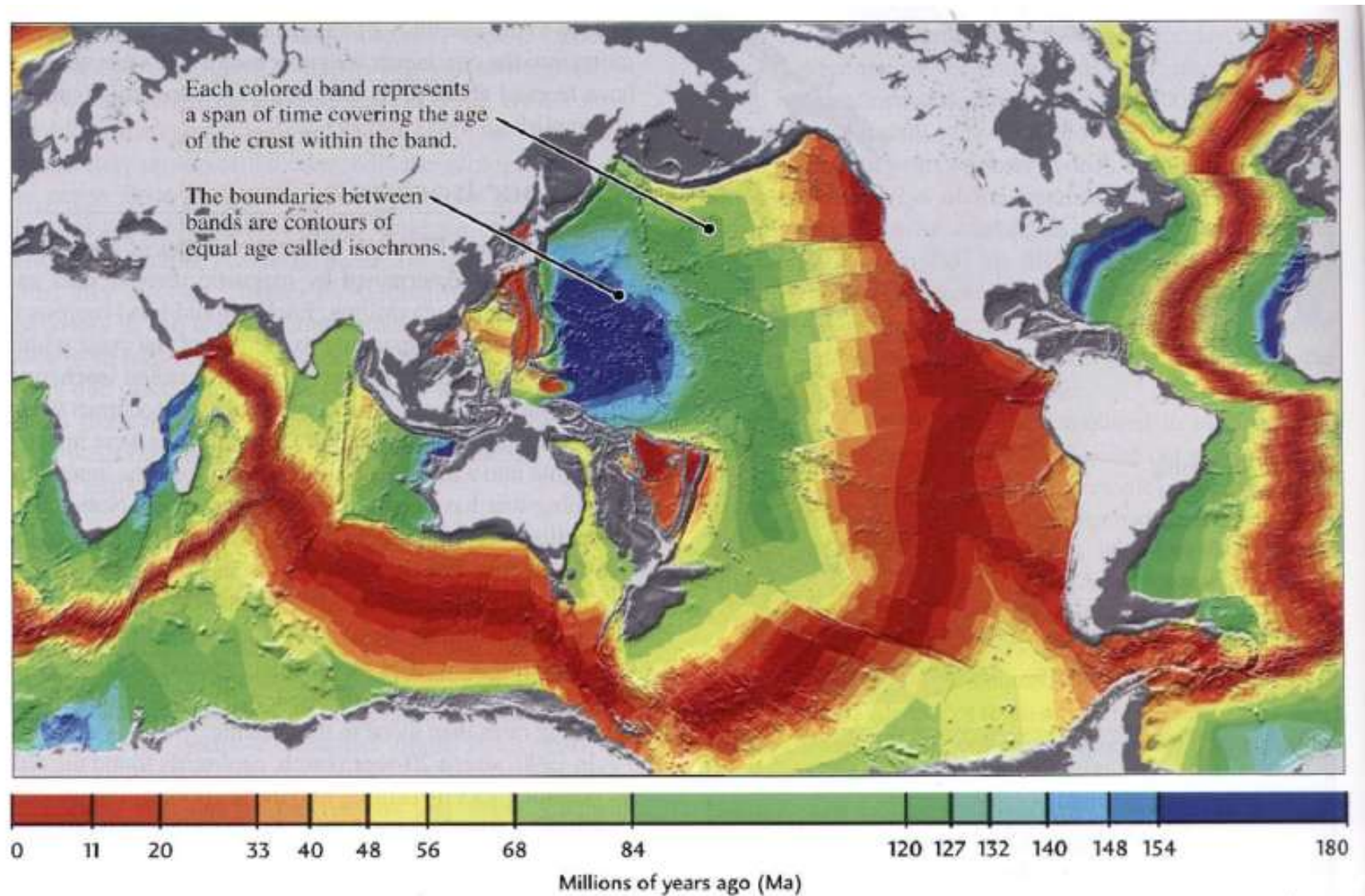
Paleomagnetism

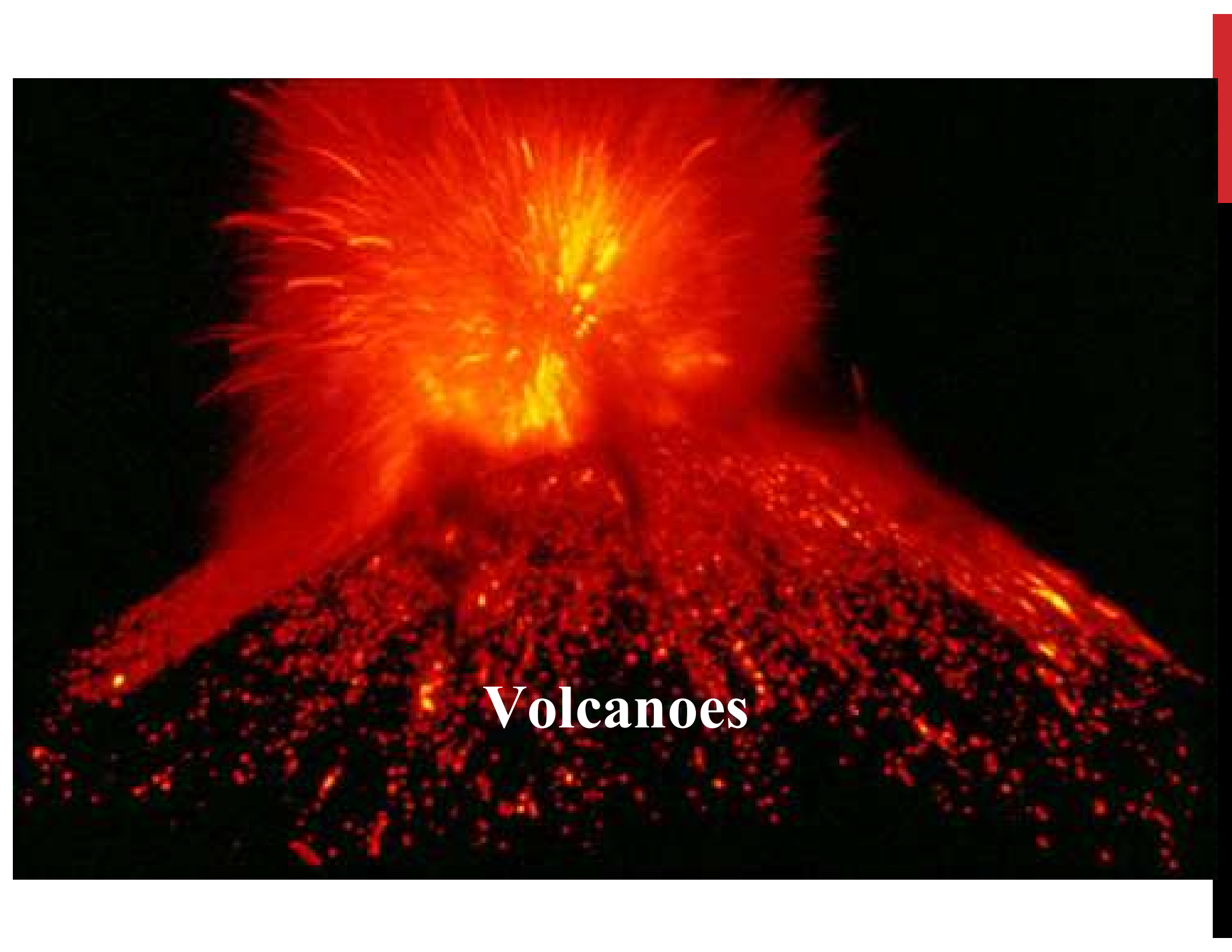


Paleomagnetism



The global isochron map of the ocean floor



A photograph of a volcanic eruption at night. A large, intense plume of fire and ash rises from a volcano, illuminated from within, creating a bright orange and yellow core. The plume spreads out at the top, with many smaller, glowing fragments of lava or ash falling back down, creating a dense, sparkling effect. The background is dark, making the eruption stand out.

Volcanoes

Eruption at Pompeii

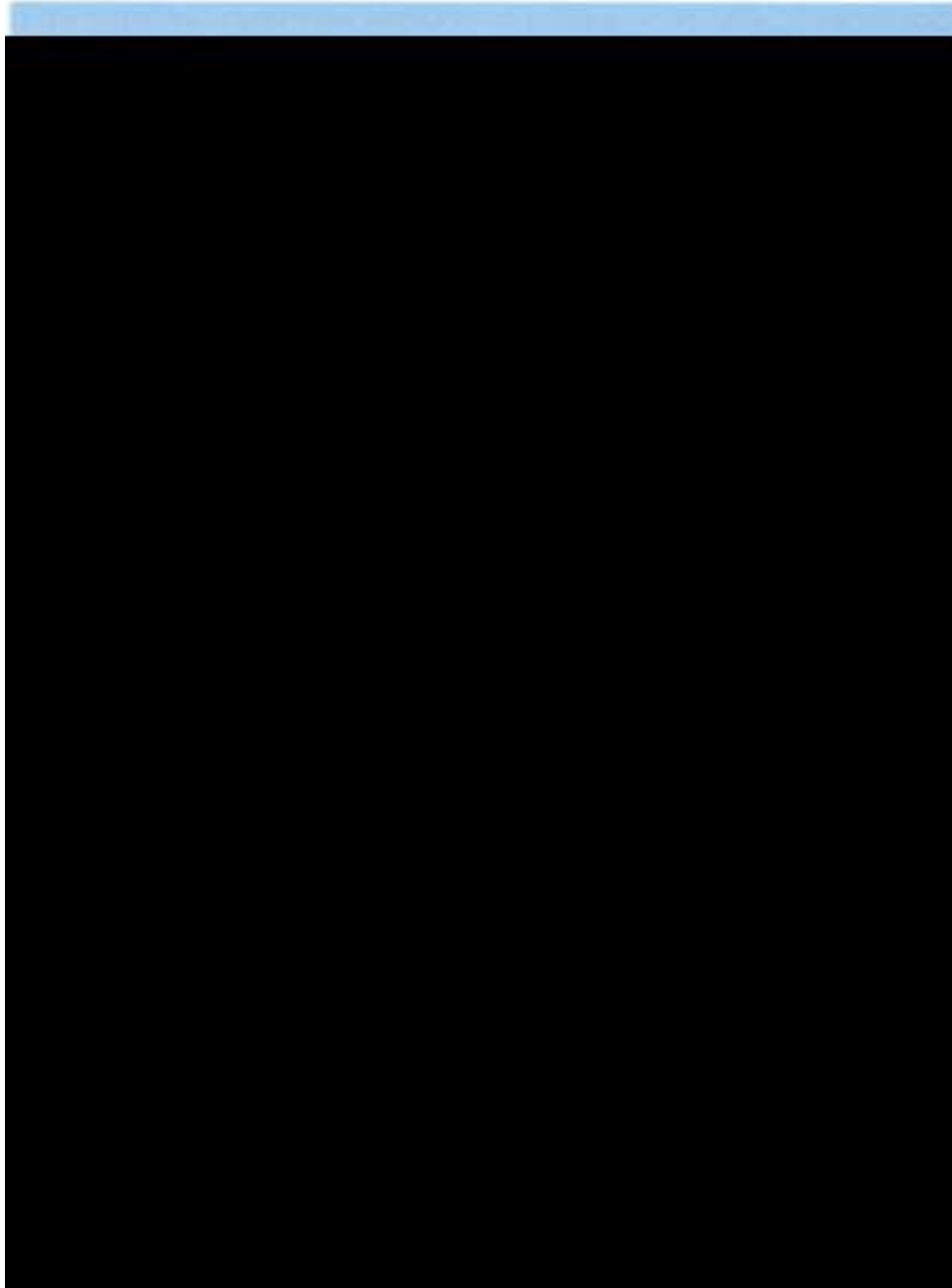


MAY 1990 ERUPTION OF KILAUEA, HAWAII



James Cachero/Sygma

Plumbing System of a Volcano



Lava accumulates on the surface to form volcano

Lava erupts through a Central and side vents

Rise through the lithosphere

Originates from the Asthenosphere

Types of Lavas

- Basaltic Lavas
- Andesitic Lavas
- Rhyolitic Lavas

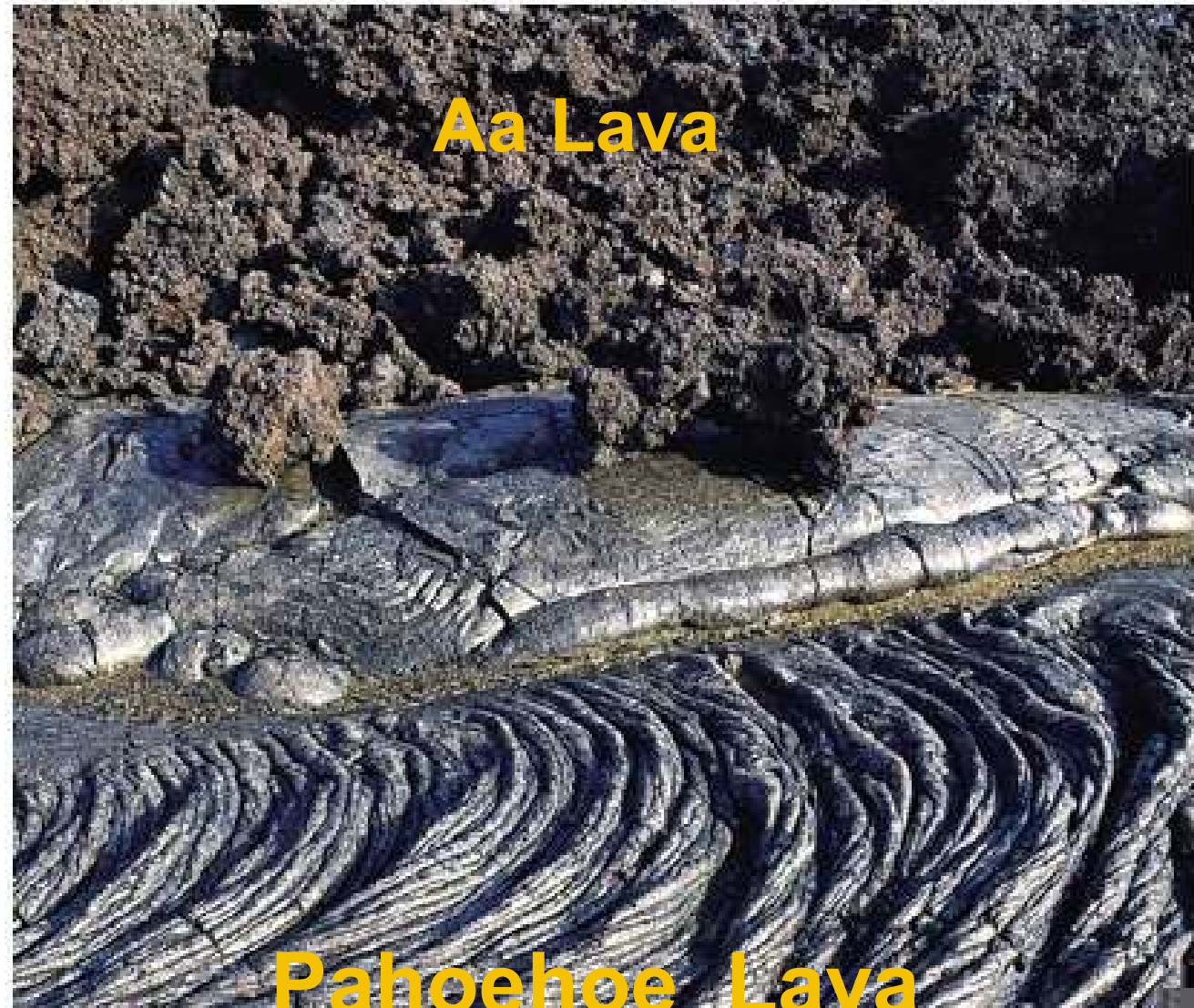
Basaltic Lavas

- Mid oceanic ridges
- Mafic in composition
- Higher temperature
- Rarely explosive

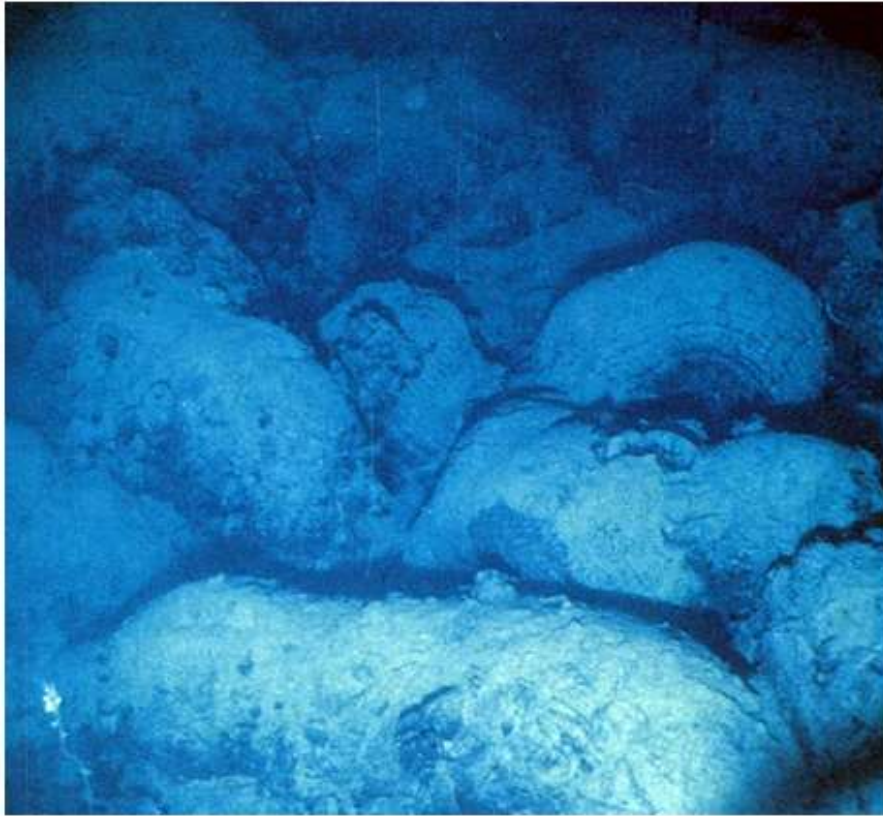


A central vent eruption from Kilauea, Hawaii produces fast flowing Basaltic lavas

Type of Basaltic Lavas



Pillow Lava



Andesitic Lavas

- Intermediate SiO_2
- Flow relatively slow
- Temperature lower
- Subduction settings
- Phreatic explosions



Mount St Helens, a andesitic volcano in south-western Washington state

Phreatic explosions



Rhyolitic Lavas

- High silica (~68%)
- Temperature lower (600-800)
- Thick bulbous deposits
- Gases are easily trapped with
in rhyolitic lavas



Viscous rhyolite lava flow. Wilson Butte dome, Long Valley, California

Textures of Volcanic rocks

Obsidian

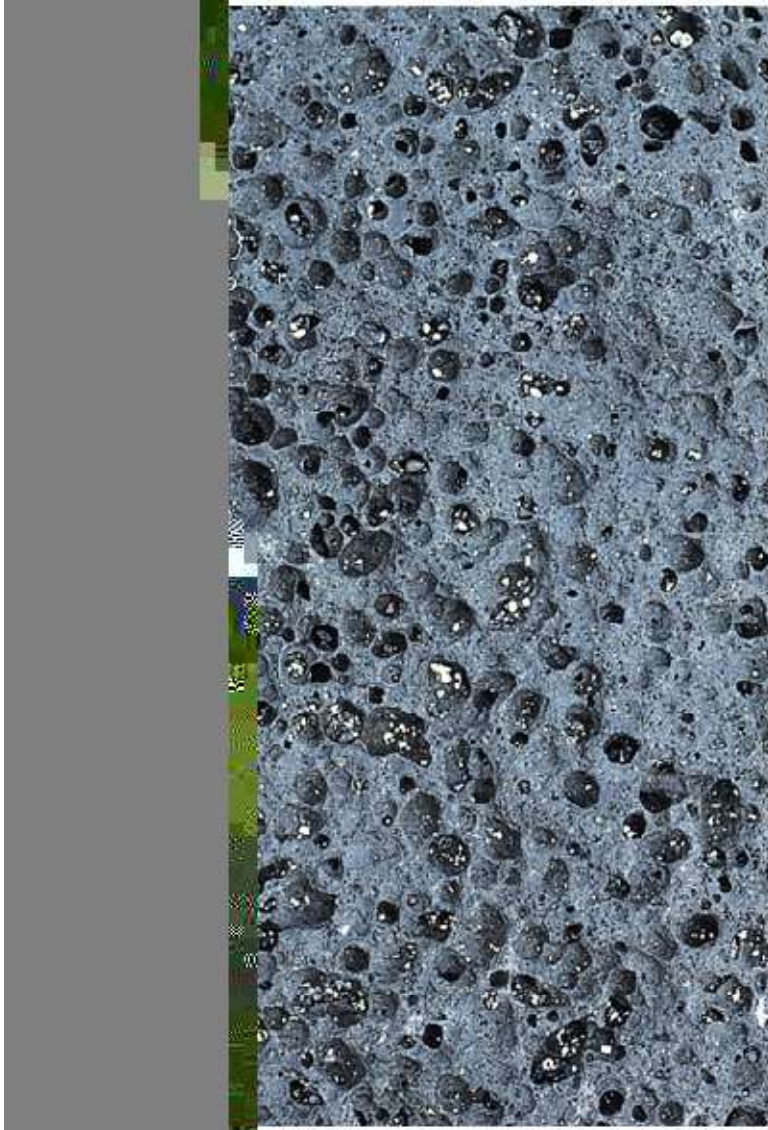
Vesicular

Obsidian



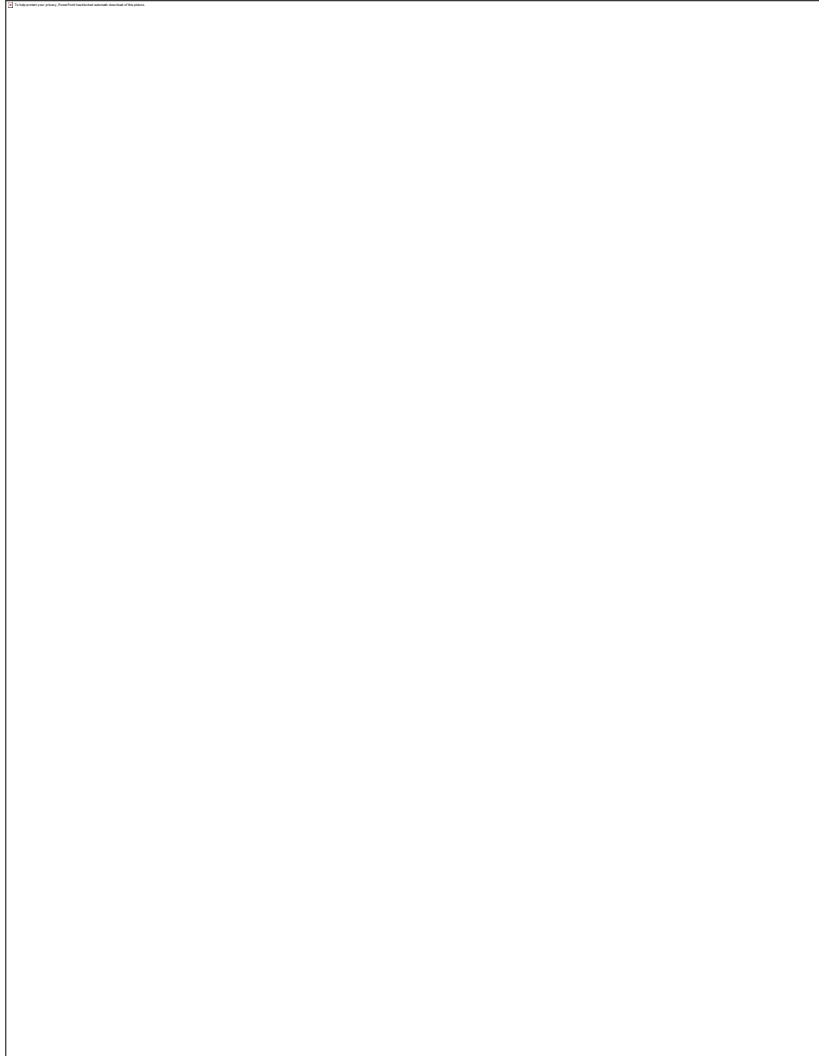
Silica rich rapidly cooled lavas form obsidian

Vesicular Basalt



Pumice floating in water

Pyroclastic deposits



Volcanic ejecta

Pyroclastic flows

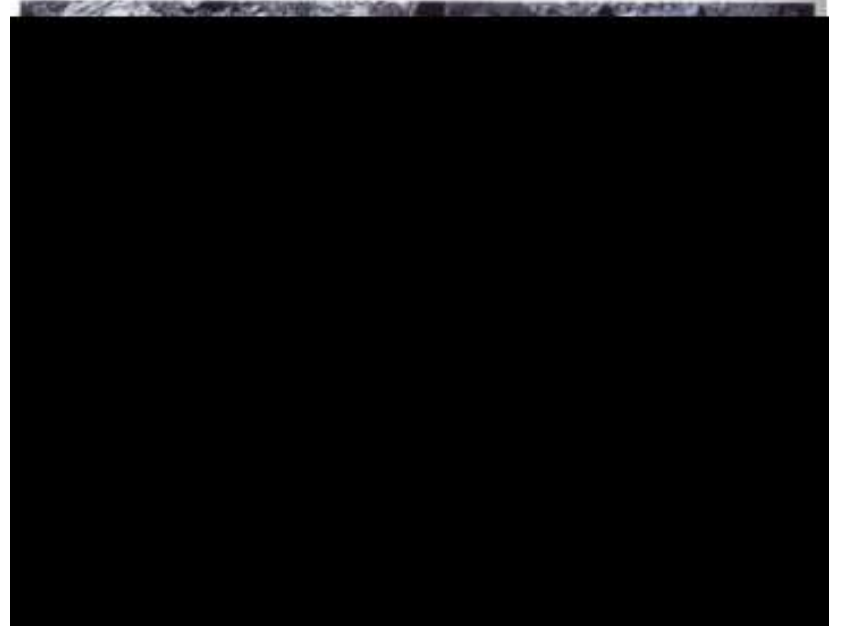
**Explosive eruption at Arenal Volcano,
Costa Rica**

Pyroclastic deposits

- **Volcanic ash** – less than 2mm in diameter
- **Volcanic Bombs**- large fragments



Volcanic ash

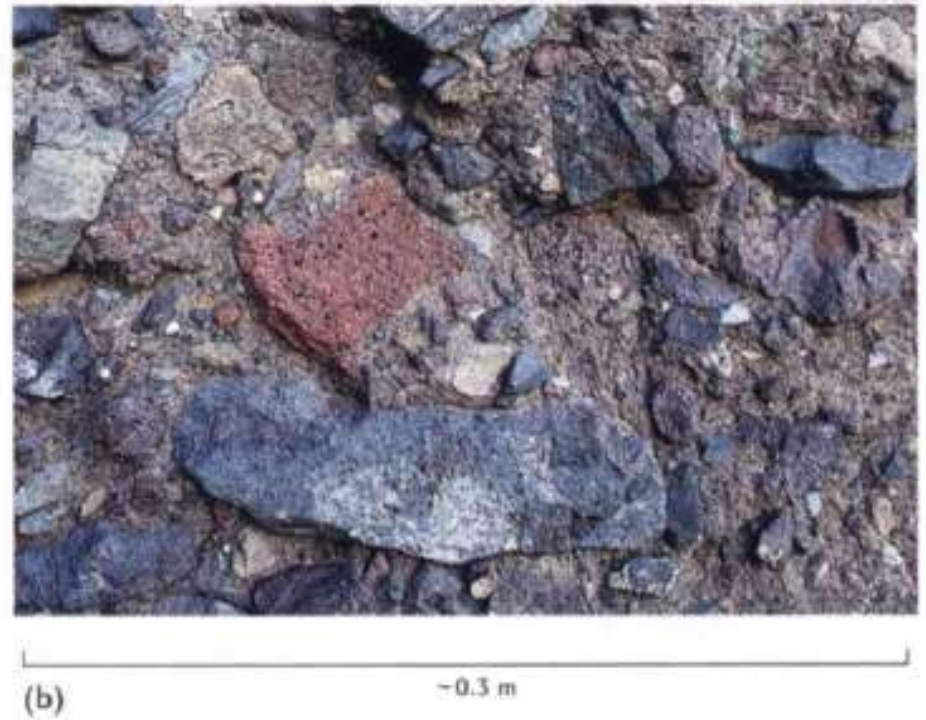


Volcanic Bombs

Pyroclastic deposits



Tuff



Volcanic breccia

Pyroclastic flow



Pyroclastic flow plunged down the slopes of Mount Unzen, Japan

Eruptive styles and landforms

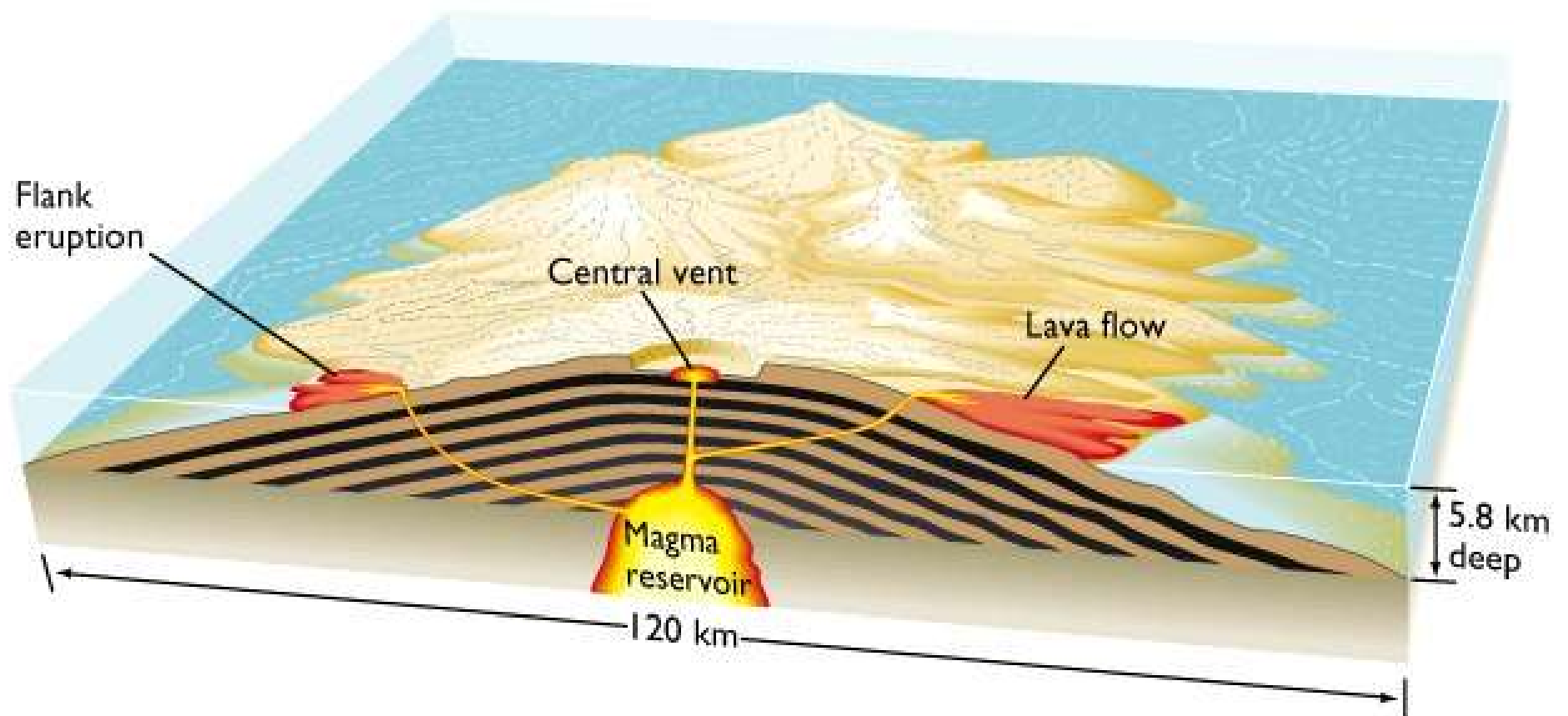
- **Composition of magma**
 - gas content
 - chemical composition
- **Type of material erupted (lava versus pyroclasts)**
- **Environmental condition in which it erupts**
(submarine vs. continental)

Central Eruptions

Shield Volcanoes

- **Low-viscosity lava flows (Mafic)**
- **Gently sloping flanks — between 2 and 10 degrees**

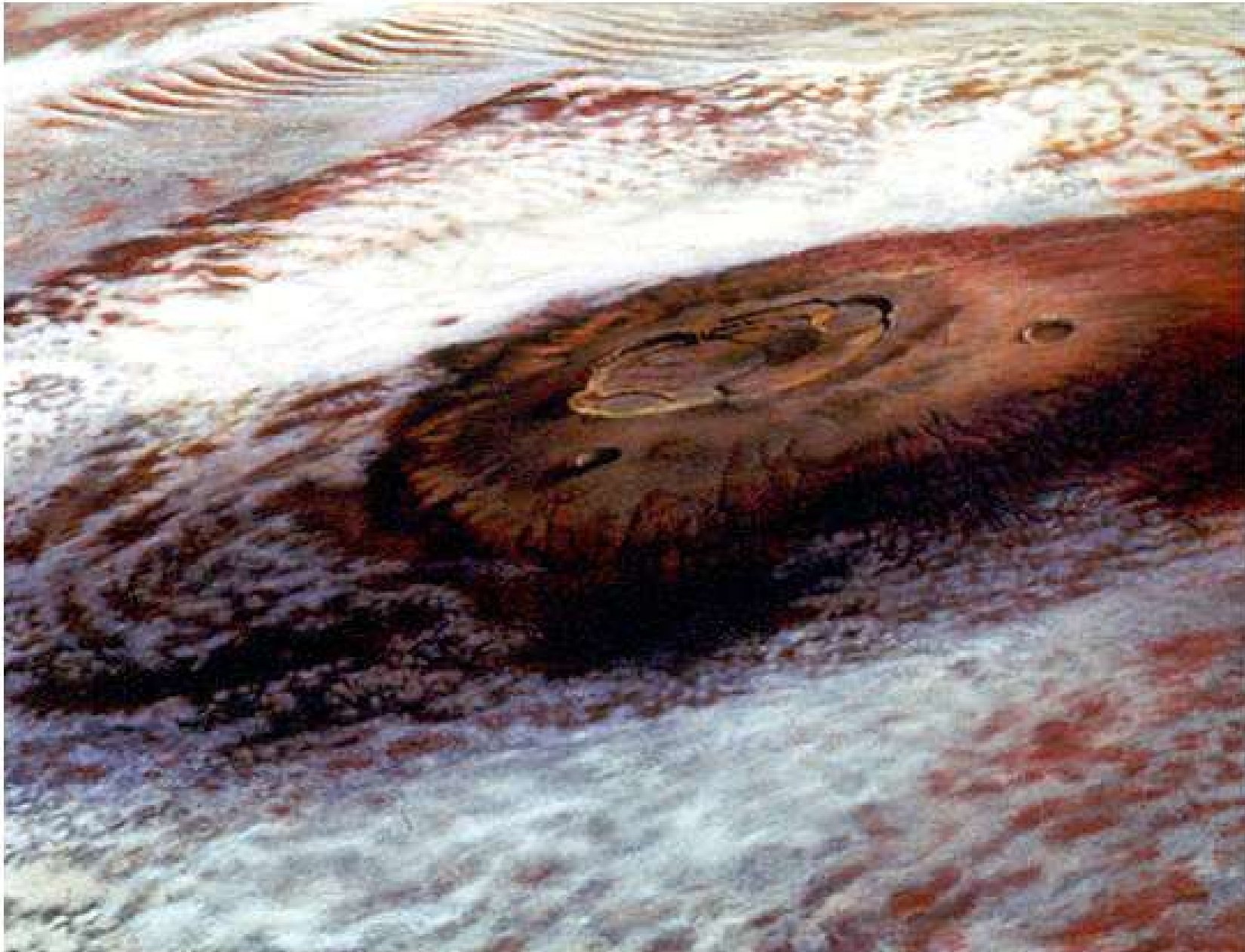
Shield Volcanoes



Shield Volcanoes



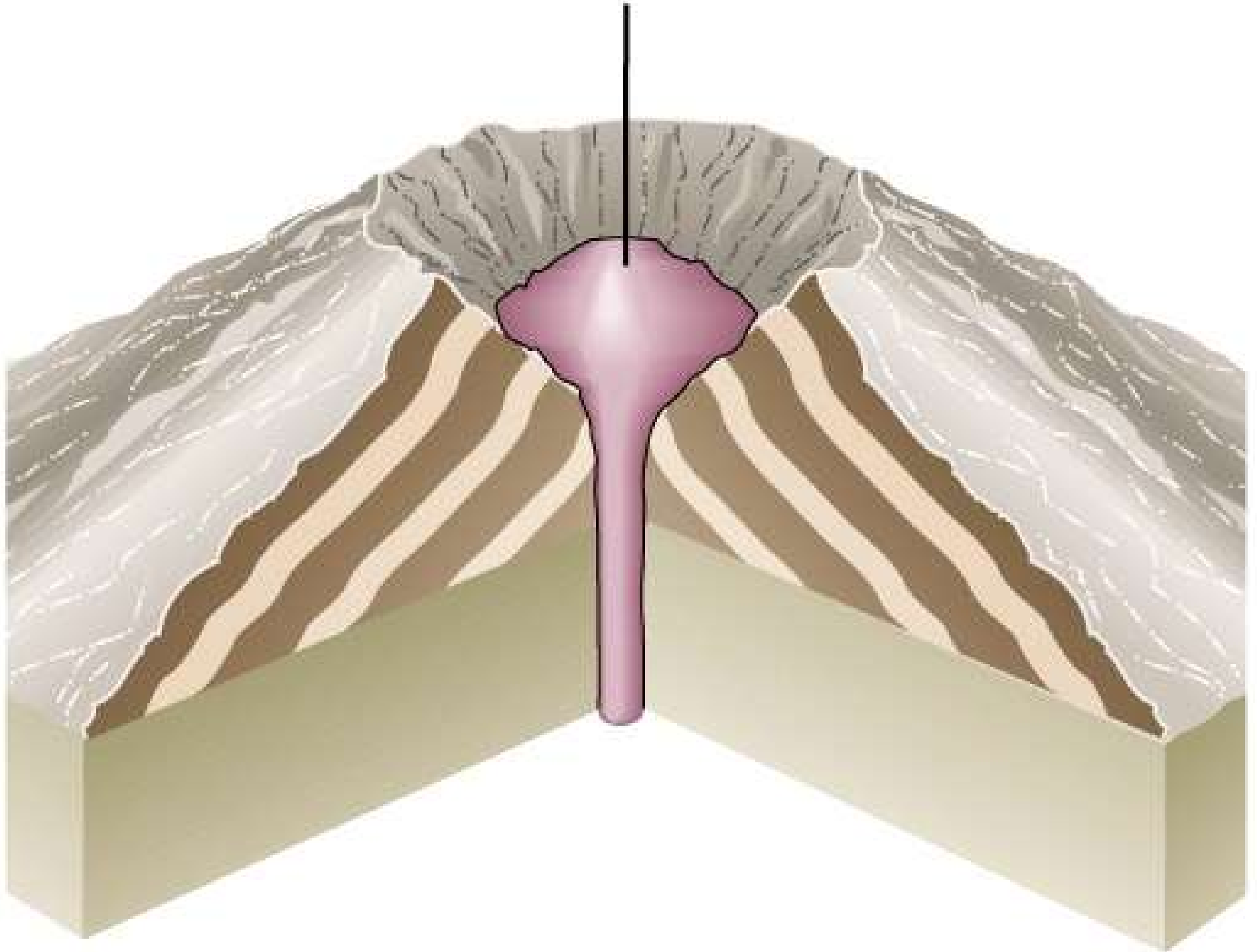
Olympus Mons Shield Volcano



Volcanic domes

- **Forms above a volcanic vent, a bulbous step sided mass of rock**
- **Viscous lava — usually silica-rich (or cooler magma)**
- **Associated with violent eruptions**

Lava dome



Lava Dome



Lyn Topinka/USGS

INYO OBSIDIAN DOMES-CALIFORNIA

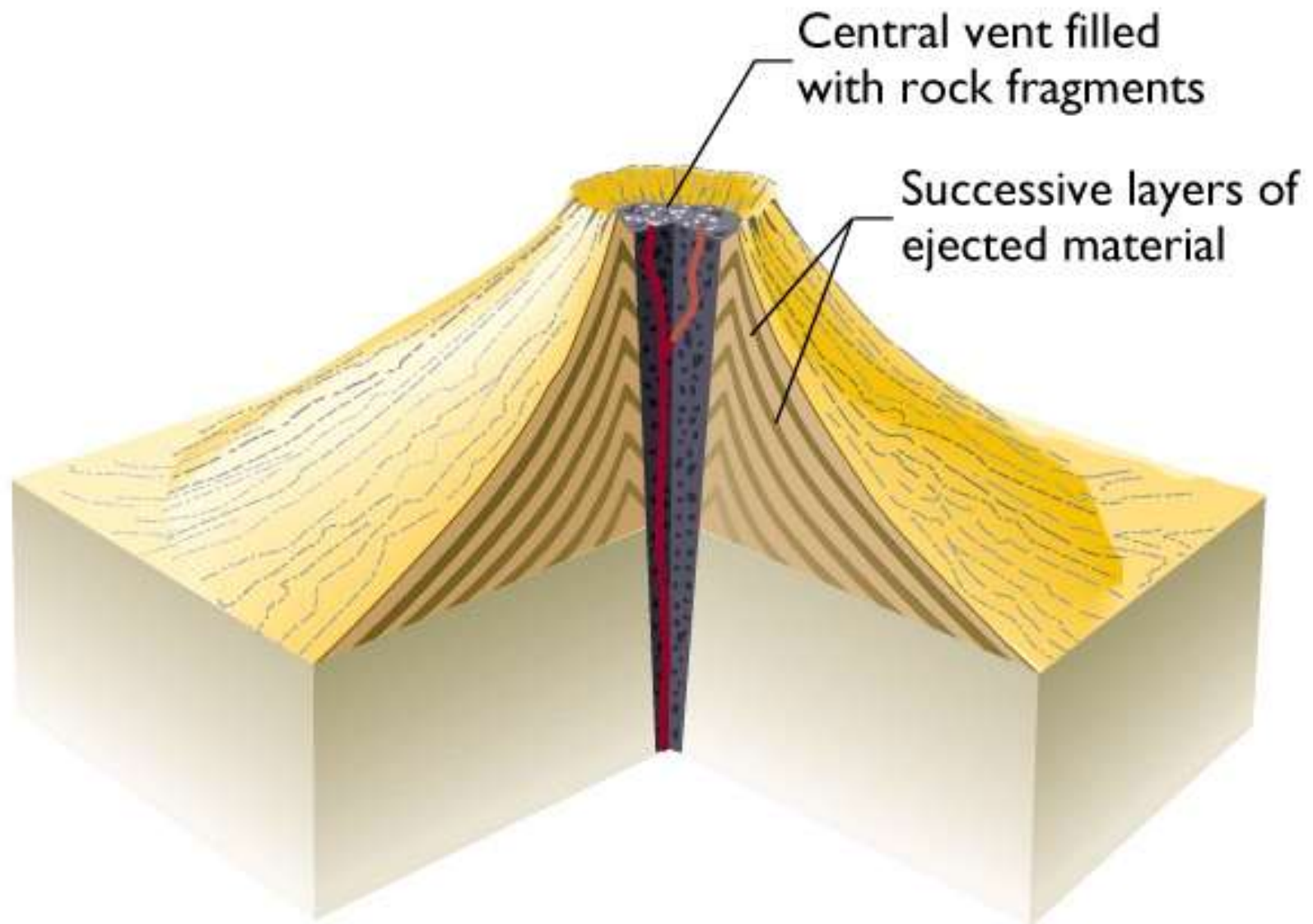


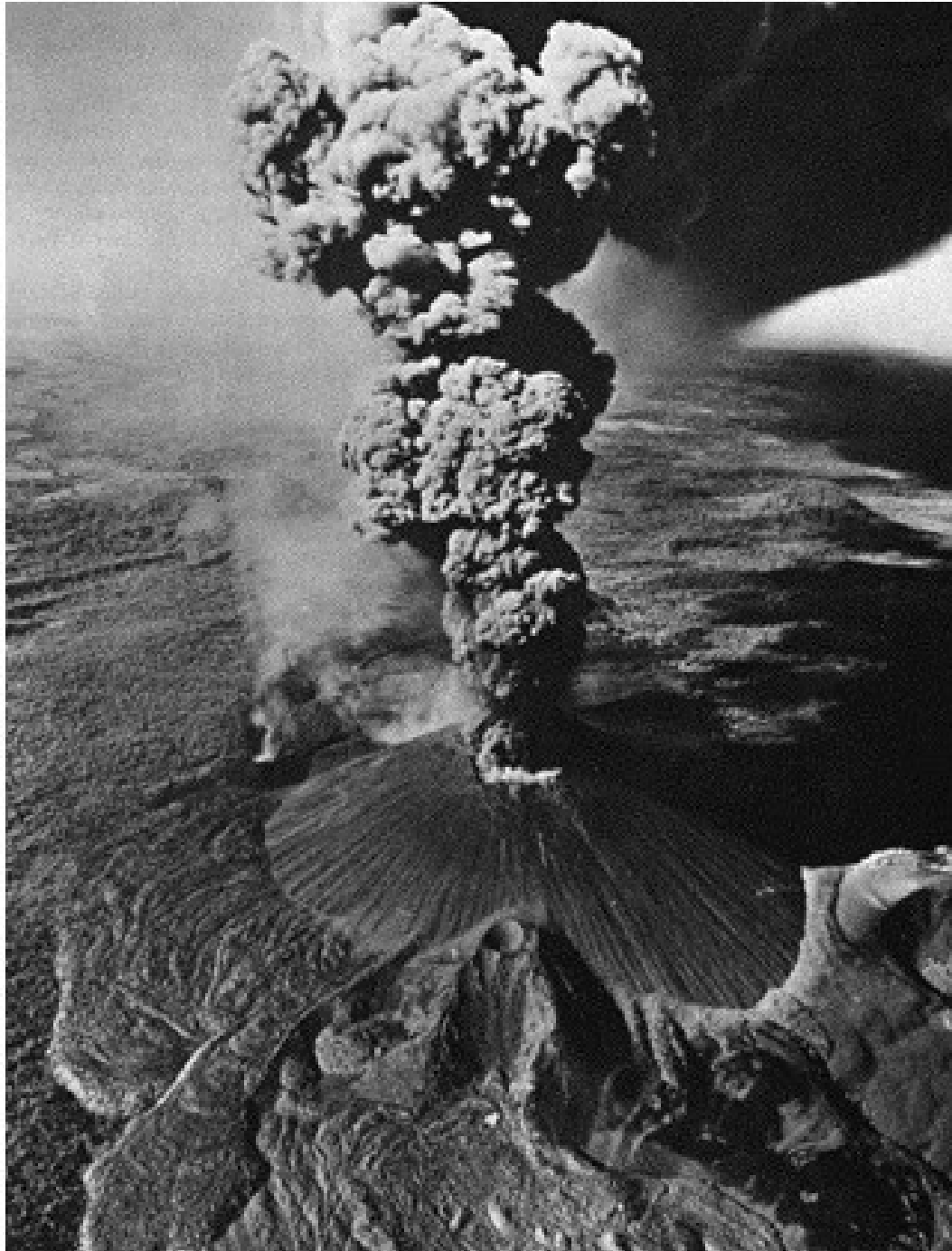
Central Eruptions

Cinder Cones

- **Formed of pyroclastics only**
- **Steep sides — ~30 degrees**
- **Short duration of activity**
- **Larger fragments near to vent form steep slopes**

Cinder Cones



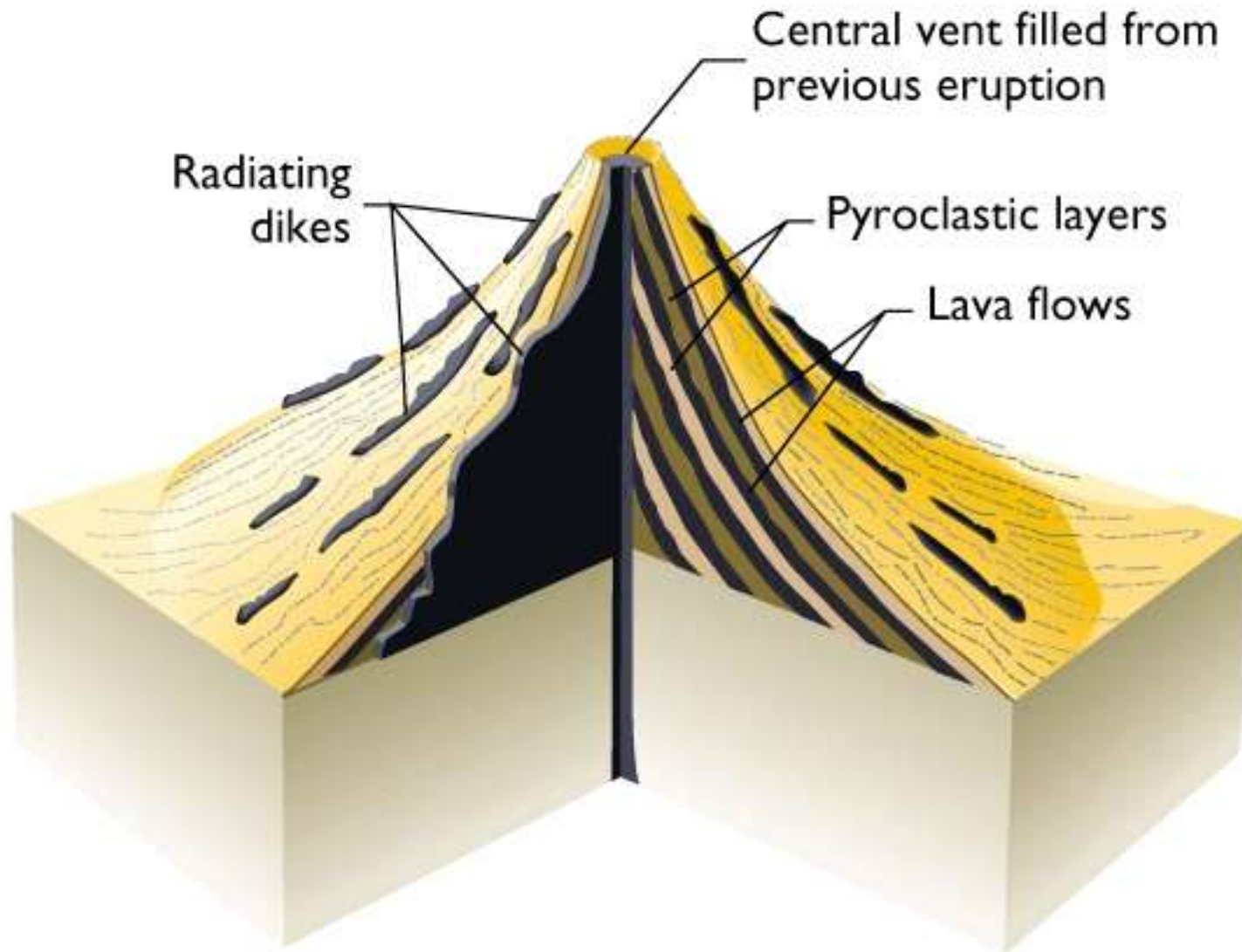


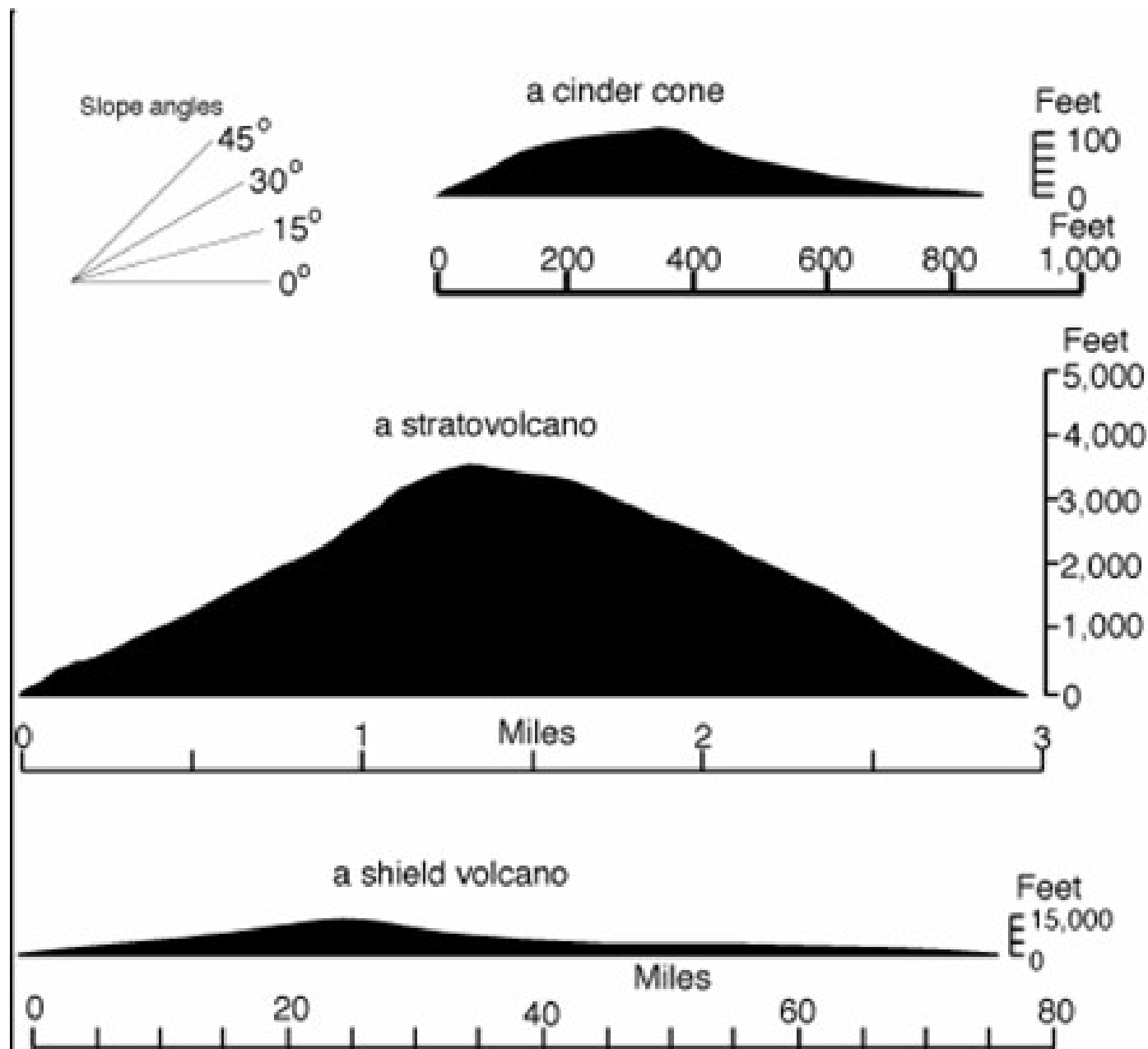
**Cerro Negro Cinder
Cone near Managua
Nicaragua in 1968**

Composite/strato Volcano

- **Alternating pyroclastic layers and lava flows**
- **Slopes intermediate in steepness**
- **Intermittent eruptions over long time span**
- **Mostly *Andesite***
- **Distribution**
 - Circum-Pacific Belt (“Ring of Fire”)
 - Mediterranean Belt

Composite Volcano





MT FUJIYAMA, JAPAN



Raga/The Stock Market

Fig. 5.15

Calderas

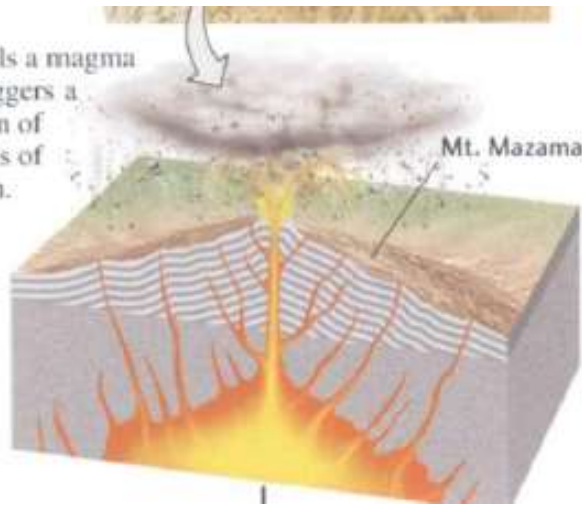
- **Depression at top of volcano produced during an eruption**
- **Great volume of magma are discharged rapidly from a large magma chamber, the chamber can no longer support it**

Calderas

Volcanic eruption

STAGE 1

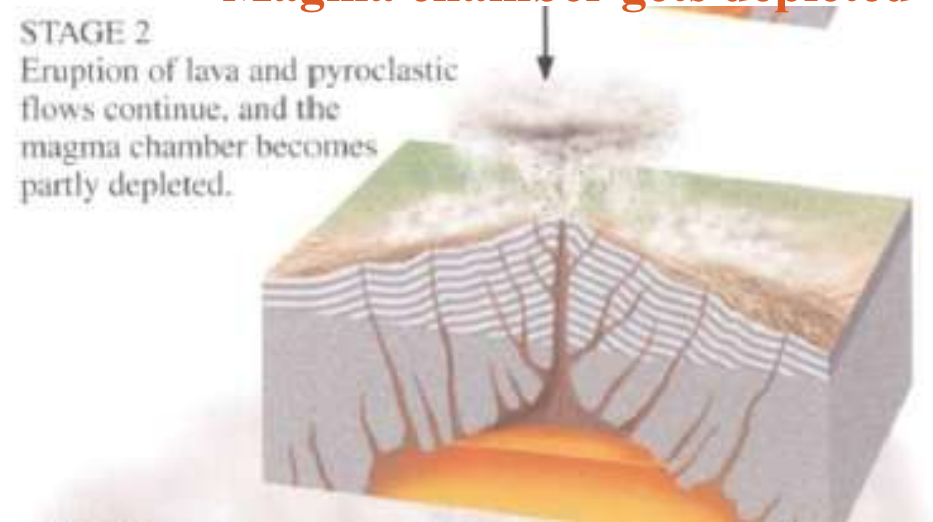
Fresh magma fills a magma chamber and triggers a volcanic eruption of lava and columns of incandescent ash.



Magma chamber gets depleted

STAGE 2

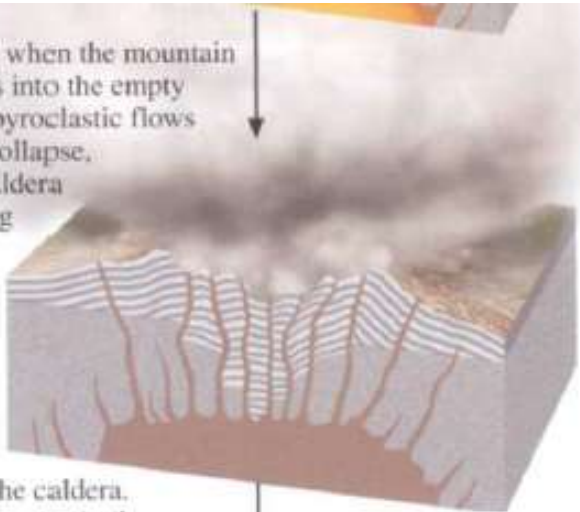
Eruption of lava and pyroclastic flows continue, and the magma chamber becomes partly depleted.



Collapse of mountain summit into empty chamber

STAGE 3

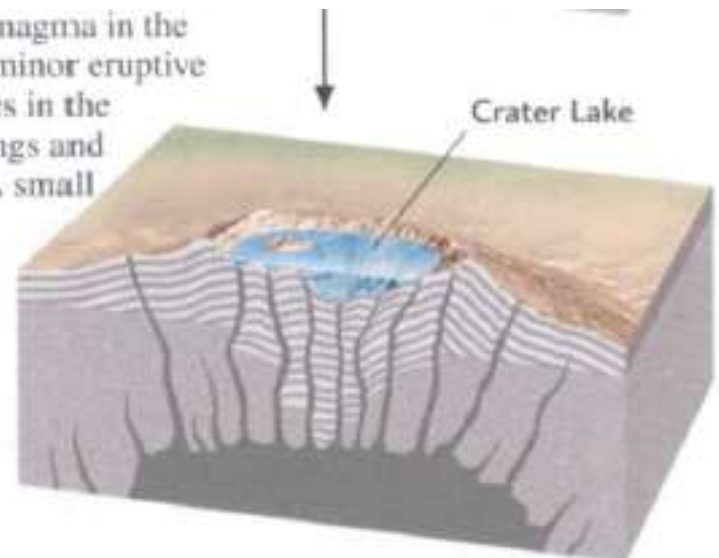
A caldera results when the mountain summit collapses into the empty chamber. Large pyroclastic flows accompany the collapse, blanketing the caldera and a surrounding area of hundreds of square kilometers.



STAGE 4

A lake forms in the caldera.

Small amounts of magma in the dikes, minor eruptive activity continues in the springs and vents. A small lake forms in the caldera.



Mt Fujiyama, Japan

Before May, 1980



I Muench/Photo Researchers

After May, 1980



David Weintraub/Photo Researchers

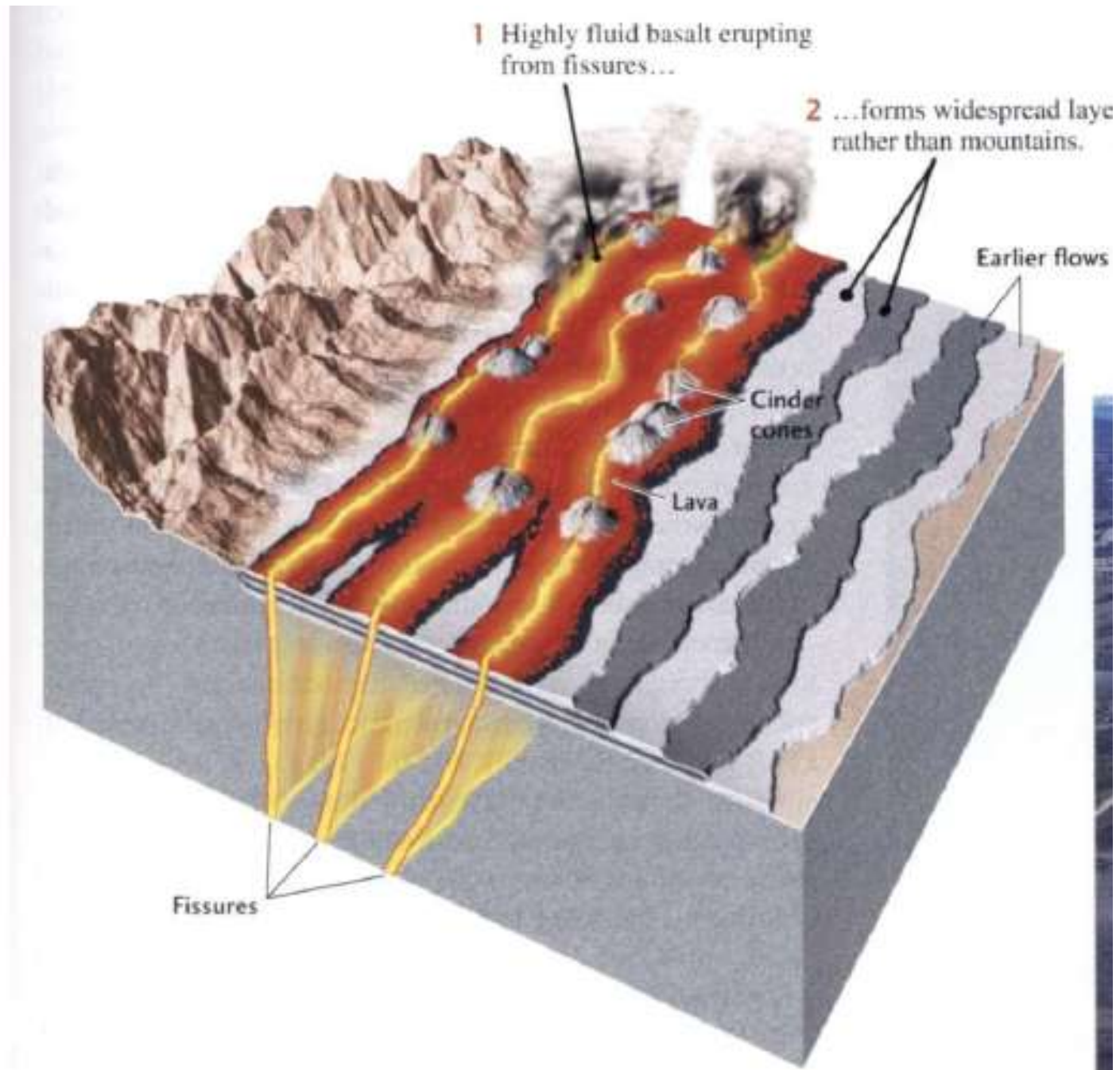
Crater

- Bowl shape pit found at the summit of volcanic mountains



Xico volcanic crater, Mexico city

Fissure type eruption



Fissure type eruption

Flood basalts



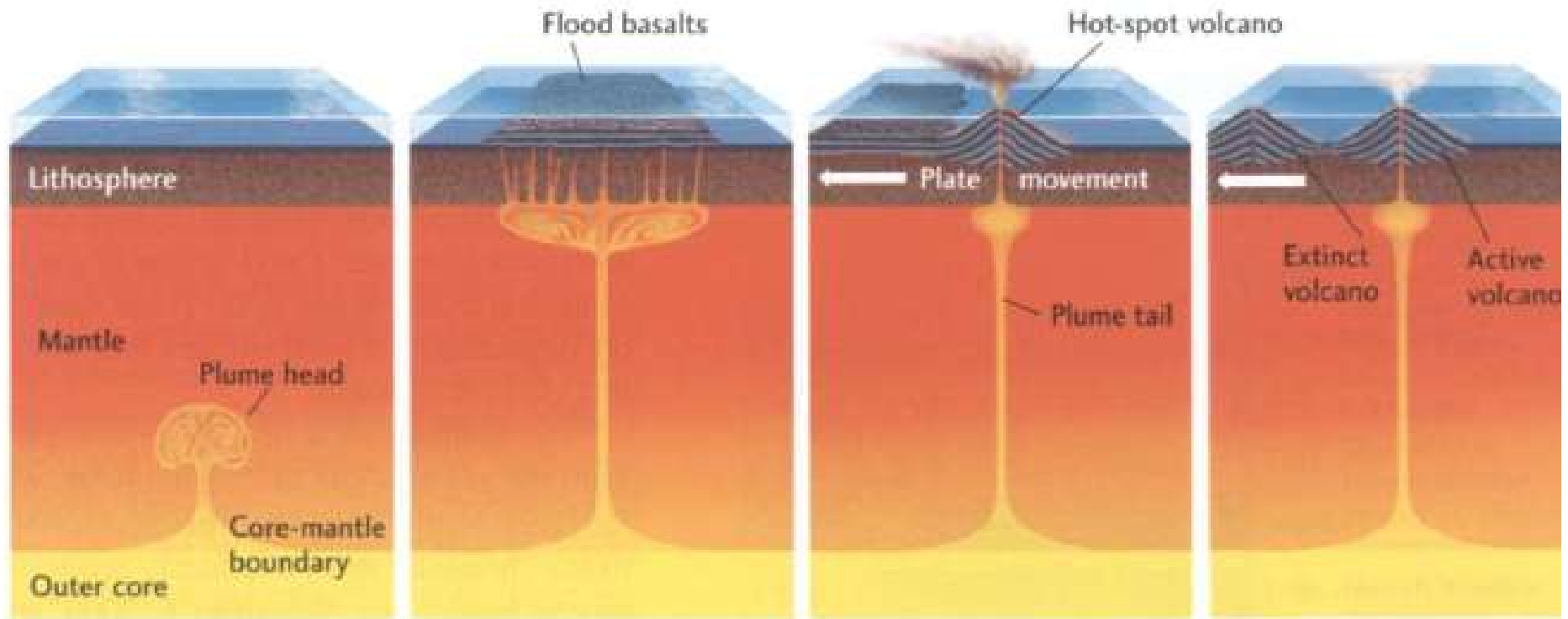
Deccan basalts

The tectonic setting of Volcanoes

- **Convergent plate boundaries**
- **Divergent plate boundaries**
- **Within plate “hotspots”**

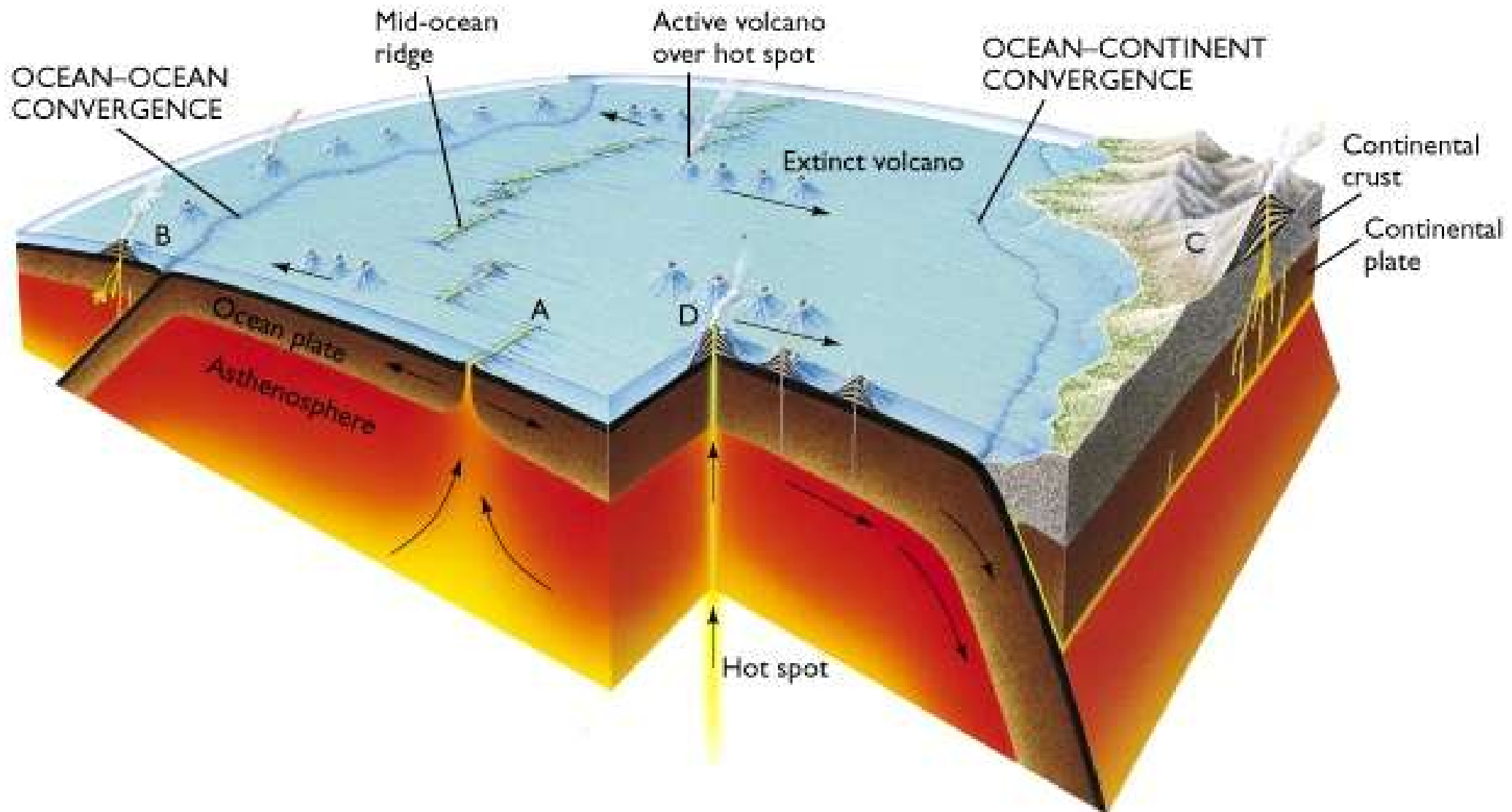
Hotspots/Mantle plumes

- Arises in the core mantle boundary
- Large turbulent blob of hot materials- 'plume head'
- At top of the mantle the plume head generates large amount of magma

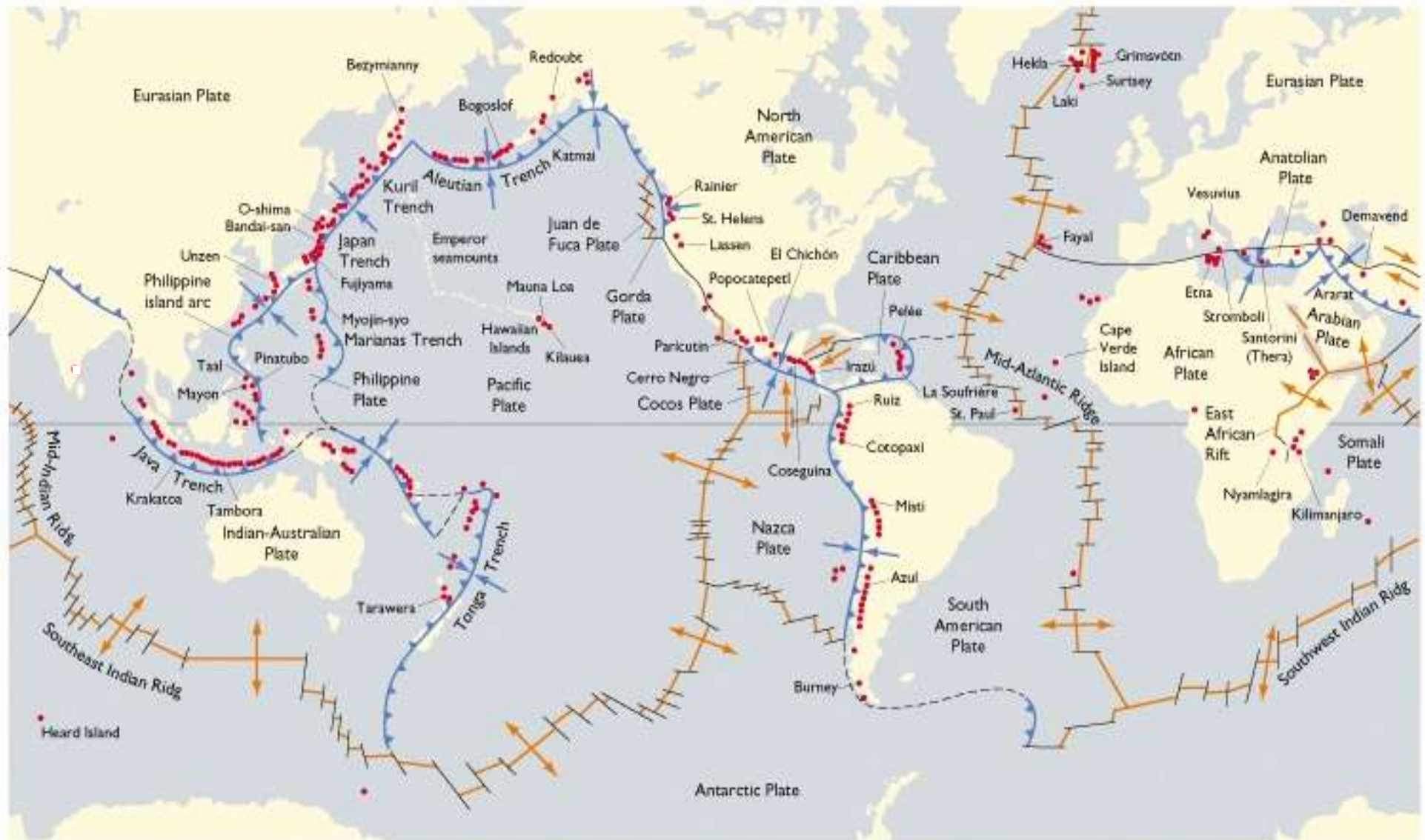


Speculative model for the formation of Hot spots

The Volcanism associated with plate tectonics



The world's active Volcanoes



PREDICTING ERUPTIONS

Why try to predict eruptions?

- Minimize damage
- Minimize loss of life

Four ways to predict an eruption:

1) Harmonic tremors

- Small earthquakes
- From moving magma

2) Increased gas emissions

- CO₂
- SO₂
- H₂S
- Large tracts of healthy forests die off

PREDICTING ERUPTIONS

3) Changes in mountain shape

- Pressure from the magma deforms the mountain
- Detected by tiltmeters

4) Observe material from past eruptions

- Geologists map out past:
 - Lahars
 - Lava flows
- Count number of past eruptions
- Date each eruption
- Calculate periodicity

PREDICTING ERUPTIONS

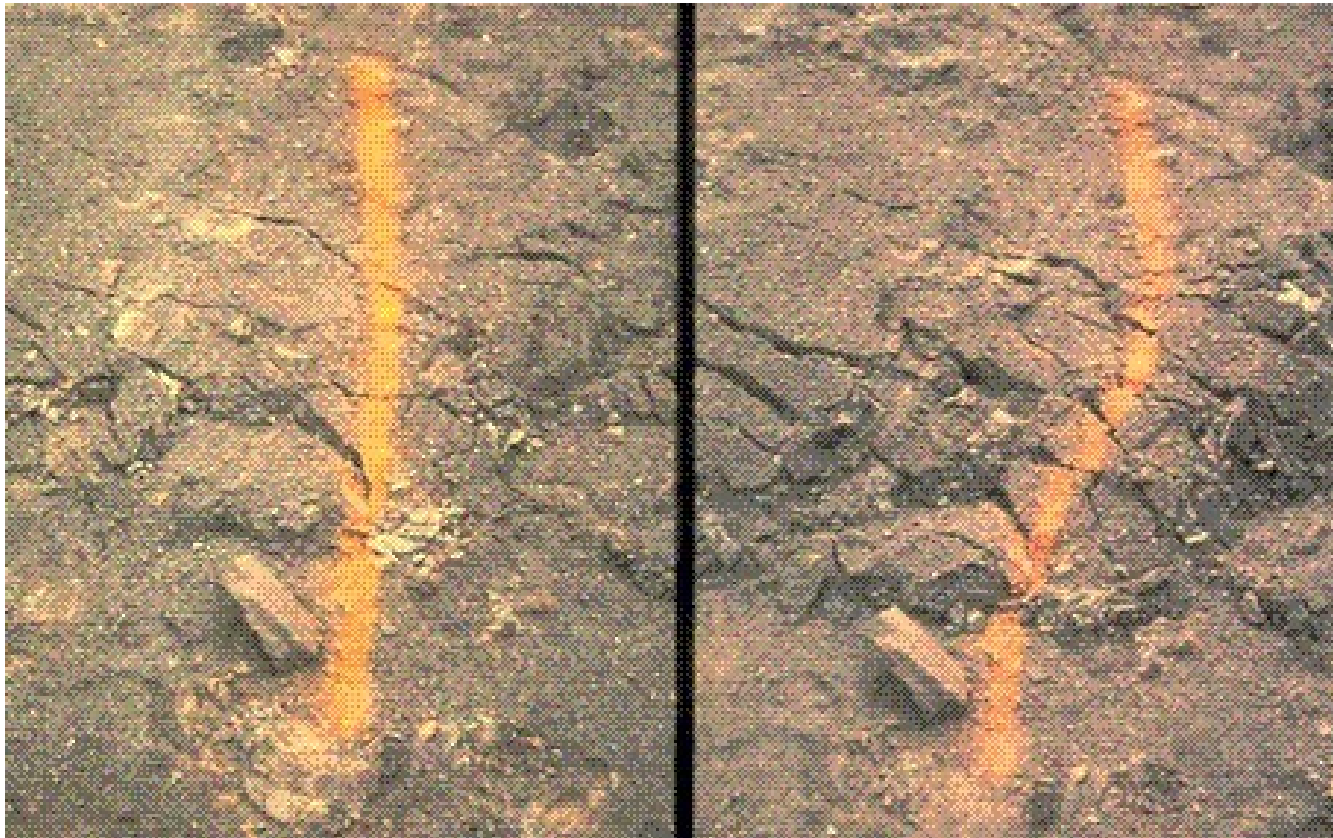
Tiltmeter

- Tiltmeter measures ground tilt
- Stable ground = zero tilt
- Change from zero indicates shape change



PREDICTING ERUPTIONS

Mt. St. Helens and the orange line



BENEFITS OF VOLCANOES

Benefits of volcanoes

- Soils
- Energy from heat
- Gases for industry